

🌀 M002 Week 1 — While Loop

Topic: Repeat Until a Wall · **Course:** Maze Madness · **Level:** Advanced · **Time:** about 55 minutes

A while loop checks, then runs the body, then checks again. It stops when the check is false.

1 · Trace the Loop 🧠

```
while not agent.detect(BLOCK, FORWARD):
    agent.move(FORWARD, 1)
```

The wall is 5 steps ahead. How many moves? When does the loop end?

2 · Read the API

Call	What it does
<code>agent.detect(BLOCK, FORWARD)</code>	true if a wall is ahead
<code>agent.move(FORWARD, 1)</code>	step forward once
<code>agent.turn(RIGHT_TURN)</code>	turn right

Why check `not agent.detect(...)` and not `agent.detect(...)` ?

3 · Spot the Bugs 🐛

Bug A — should move until the wall.

```
while agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

Fix it:

Bug B — should stop right at the wall, not crash.

```
while not agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)  
    agent.move(FORWARD, 1)
```

Fix it. Say why it crashed:

4 · Write a Chat Command

Make `go` move the agent forward until a wall.

Use `player.on_chat` , `while` , `agent.detect` , `agent.move` .

5 · Loop, Turn, Loop

Move to the wall, turn right, then move to the next wall.

6 · Finish the Maze 📷

Open your homework world. Use a while loop to drive the agent down the straight maze to the wall.

Send a photo OR video of the agent at the wall. Then write 4-6 sentences.

First, I read the maze and saw ...

My loop checked ...

The agent stopped because ...

The hard part was ...

A while loop beats many **move** lines because ...

7 · Narrate Your Walkthrough 📹

Record on a phone while the agent runs. Teach it like the viewer is new. Use: **while**, **loop**, **detect**, **wall**, **forward**.

1. Show the start. Run your code.
2. Read your loop. Say what makes it stop.
3. Show one bug and your fix.
4. Say why a loop beats many `move` lines.

Plan your words:

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.